AMIT SINGH RAJAWAT

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EDUCATION

Madhav Institute of Technology & Science, Gwalior — CGPA: 8.65
B.Tech in Information Technology (Artificial Intelligence & Robotics) — *Aug. 2020 – May 2024*

WORK EXPERIENCE

Software Engineer (AI/ML) — June 2024 – Present
Kloudspot Inc. — *Bengaluru, India*

- Designed a **Face Recognition Algorithm** for **1:1 Verification and 1:N Identification in C++**, achieving a global rank of 277 in the **NIST-Face Recognition Vendor Test (FRVT) in 1:1 Verification**.
- Trained models using **ResNet-200** architecture with **Convolutional Block Attention Module (CBAM)** and implemented face recognition techniques, improving model accuracy by 13%.
- Developed an **FRS (Facial Recognition System) Analysis Tool** from the ground up, enabling the **analysis of passenger movements** using **clustering techniques**. The tool provides **visualizations** to **track individuals** from one **gate** to another within the **airport**.
- Created the **FRS Registration Tool** from scratch using **Python**, implementing **detection**, **landmark processing**, and **recognition**, enabling accurate **multi-pose registration** and **real-time identification** and accurately **tracking individuals** in **live environments**.

Software Engineering Intern — Feb 2024 – May 2024
Kloudspot Inc. — *Bengaluru, India*

- Engineered a high-performance **computer vision inferencing framework** in **C++** using **GStreamer**, **OpenVINO**, and **DLStreamer**, resulting in a **9%** improvement in processing speed.
- Implemented **gRPC** for effective communication between client and server, along with **GStreamer RTSP Server** for seamless real-time video broadcasting which supported high-quality video transmission with minimal packet loss.
- Refined **pre-processing and post-processing workflows** to enhance the accuracy and efficiency of model predictions.

Project Intern (LOR) — Dec 2023 – Jan 2024
Indian Space Research Organisation (ISRO) — *Bengaluru, India*

- Developed a **vision-based satellite pose estimation system** using the **HRNet algorithm**, improving landmark detection for satellite orientation estimation in varying conditions, achieving 90% accuracy.
- Integrated **deep landmark regression** and **nonlinear pose refinement** techniques, training models on extensive datasets to accurately capture satellite orientations, reducing error rates by 15%.

Amazon ML Summer School — Sept 2023 – Oct 2023
Amazon — *Virtual*

- Implemented **Supervised**, **Unsupervised**, and complex **ML algorithms** using a wide range of **Python libraries**.
- **Worked** on various **machine learning projects** employing sophisticated **algorithms** and **libraries**, achieving an **accuracy** of over **95%**.

PROJECTS

Snake Game Using RL With Neural Network | *Python, TensorFlow, Keras, Matplotlib, NumPy*

- **Enhanced gameplay** using **Q-learning**, improving the agent's performance through trial and error and state-space representation, and used **boolean features** to improve decision-making and gameplay strategies by 20%.
- Visualized agent performance and evolving strategies with **animated plots** for deeper understanding, improving analysis efficiency by 10%.

Text Classification With TensorFlow | *Python, TensorFlow, Keras, Scikit-learn, pandas, NumPy*

- Created a **high-performance text classification model** for **sentiment analysis** and **document categorization**, leveraging advanced **NLP** and **DL techniques**, and enhanced model performance through **hyperparameter tuning**
- Deployed the optimized model using **TensorFlow**, demonstrating advanced **machine learning** skills in text-based applications, reducing prediction time by 15%.

Hand Gesture Recognition using OpenCV | *Python, OpenCV, TensorFlow, Keras, NumPy, Matplotlib*

- Trained a **Temporal Convolutional Network (TCN)** on 12k RGB videos to learn efficient temporal modeling in image sequences for **hand gesture recognition**, enabling downstream music control tasks such as **pause, play, and next**.
- Achieved **92% accuracy** by preprocessing images with **background removal techniques**, using **color-based segmentation** and **SIFT**.

PUBLICATIONS

- A. Rajawat, A. Manjhi and A. Srivastava, "**Categorization of IRIS Flower using Different Machine Learning Strategies,"** [paper]
- A. Manjhi, A. Rajawat and A. Srivastava, "**Design and Analysis of Programmed Face Monitoring System**," [paper]
- Rajawat, A.S., Srivastava, A. "**Recognition of Parkinson's ailment by using various machine learning procedures**". Current Psychology (2024). [paper]
- **On Heart Disease** (Communicated in Journal).

TECHNICAL SKILLS

- **Programming Languages**: C/C++ (Proficient), Python (Proficient), JavaScript, HTML/CSS, LaTeX, Shell, MySQL
- **Developer Tools and Frameworks**: Git, GitHub, Linux, Docker, VS Code, OpenVINO Toolkit, GCP, MLOPs
- **Libraries**: Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn, Flask, OpenCV, Transformers, Pytorch3D, MXNet, Keras
- **Coursework**: DBMS, Machine Learning and Optimization, Operating Systems, Probability and Random Processes, Computer Networks, Robotics, Computer Vision, NLP, AI

ACHIEVEMENTS

- **Leetcode:** 750+ Problem solved & 1765+ Rating **(Profile Link)** , **GFG:** 320+ Problem Solved **(Profile Link)**, Featured in **Patrika Newspaper** (Web Link) (Paper Cutting)
- Selected for **IIT Madras Pravartak Undergraduate Fellowship** 2023-24 **(among 72 in India) (Link)**